

Manav Patel

Software Engineer - AI Engineer - GenAI - Agentic AI

+91 7623949059 manavpatel072005@gmail.com linkedin.com/in/manav-patel github.com/Pm3949 Portfolio

Education

Indian Institute of Technology (Indian School of Mines), Dhanbad

Oct 2022 – June 2027

B.Tech. in Electronics and Communication Engineering & B.Tech in Computer Science and Engineering JH, India

CGPA: 8.49 / 10.0 | Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Probability & Statistics, Database Management Systems

Experience

Tech Mahindra May 2026 – July 2026
TechnoRise AI Engineer Intern Hyderabad, India

- Architected an on-premise AI inference pipeline for Project Orion, enabling secure, localized deployment of Large Language Models (e.g., Qwen 2.5) for enterprise applications.
- Evaluated high-throughput frameworks and deployed NVIDIA NIM for production serving, alongside a custom FastAPI-based gateway using vLLM for dynamic GPU resource management.
- Optimized hardware efficiency by engineering concurrency management, handling cold starts, and resolving FlashInfer environment dependencies without relying on rigid containerization.
- Integrated local inference servers into RAG pipelines and multi-agent frameworks (CrewAI) to drive complex, production-ready workflows.

Technical Skills

Languages:

C++, C, Python, JavaScript, TypeScript, SQL

Frontend:

React.js, Next.js, Tailwind CSS, Vite, Redux, Zustand, TanStack Query

Backend:

FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.io

AI/ML & Agents:

LangChain, LangGraph, CrewAI, RAG, Agentic AI, Multi-Agent Systems, Tool Calling, Function Calling, LLM Inference

RAG & AI Systems:

pgvector, Semantic Search, HyDE, Cross-Encoder Reranking, Vector Databases, Prompt Engineering

Databases:

PostgreSQL, MongoDB, Supabase, pgvector, SQL

Authentication & APIs:

JWT, OAuth 2.0, OAuth PKCE, RBAC, Row Level Security (RLS), API Integration, Webhooks

AI Infrastructure:

vLLM, NVIDIA NIM, Ollama, OpenAI API, Groq, Google TTS, Gemini API

Systems & Tools:

Linux, xv6-riscv, Git, GitHub, Postman, Docker

Achievements

- Secured an **All India Rank of 8,485** among 1,50,000 candidates in the JEE Advanced 2022 examination.
- Secured an **All India Rank of 9,294** among 12 million candidates in the JEE Mains 2022 examination.
- Achieved a **global rank of 1328** in LeetCode Weekly Contest 466 among thousands of participants.
- Solved over **900 DSA problems** across platforms including LeetCode, Codeforces, and CodeChef.

Projects

BlinkBot | Demo

React, Redux, FastAPI, Redis, API Gateway, LangChain, LangGraph, PostgreSQL, pgvector, WebSockets

- Engineered an enterprise-grade **AI Agent builder platform** using **FastAPI and LangGraph**, enabling the deployment of custom **multi-agent networks** and **supervisor-delegated routing** over private documents.
- Built an advanced **RAG pipeline** with **pgvector, HyDE, and Cross-Encoder Reranking**, supporting multi-source ingestion and real-time web search for contextual retrieval.
- Implemented **LLM tool calling** with a custom **API Manager** and **Human-in-the-Loop (HITL) WebSocket interrupts**, ensuring safe execution of dynamic webhooks and database operations.
- Designed a secure **multi-tenant architecture** with **Supabase RLS**, integrating an automated decoupled billing system for **token-based credit deductions** across concurrent workspaces.
- Deployed **Redux-based state management** and an **API Gateway with Redis caching** for centralized request handling, reducing redundant **database hits by 40–50%** and improving overall application performance.

SubjectHub | Demo

Next.js, TypeScript, Node.js, Express.js, MongoDB, Socket.IO, JWT

- Engineered **SubjectHub**, a full-stack academic collaboration platform using **Next.js, Express.js, and MongoDB** for subject-centric communication and resource management.
- Implemented **real-time chat, threaded discussions, notifications, and presence tracking** using **Socket.IO**, enabling live collaboration within subject channels.
- Built modular **REST APIs** for subjects, announcements, events, messages, resources, and users, with **JWT authentication** and role-based authorization for students, professors, and class representatives.
- Developed academic workflow features including resource uploads, announcements with attachments, calendar events, membership management, and personalized notifications.

xv6-riscv: Custom System Call Extensions

C, RISC-V, Operating Systems

- Collaborated in a team of five to extend the **xv6-riscv kernel** with custom system calls for IPC, process management, synchronization, signaling, and system monitoring.
- Engineered a **blocking IPC mechanism** using per-process kernel mailboxes, **copyin/copyout**, sleep/wakeup primitives, and spinlocks for safe concurrent communication.
- Implemented **kernel-level mutexes, process signaling, and getppid**, working across process tables, system-call dispatch, scheduling, and trap handling.
- Built dedicated **user-space test utilities** to validate concurrency, process lifecycle, inter-process communication, and system telemetry.

HectoClash | Demo

React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.io, JWT

- Collaborated in a team during a **36-hour Hackfest'25** at IIT ISM Dhanbad to build a responsive full-stack web app.
- Engineered **HectoClash**, a real-time multiplayer mental math game inspired by the Hectoc puzzle format.
- Implemented **live matchmaking and gameplay logic** enabling time-based competitive challenges.
- Designed an interactive **leaderboard, user profiles, and performance stats** to enhance user engagement.